Fiscal Year:	FY 2026	**Task Last Updated:**	FY 10/09/2025
PI Name:	McLean, Robert Ph.D.		
Project Title:	Polymicrobial Biofilm Growth and Control during Spaceflight		
Division Name:	Physical Sciences, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	McLean@txstate.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	5125581509
Organization Name:	Texas State University-San Marcos		
PI Address 1:	Biology		
PI Address 2:	601 University Dr		
PI Web Page:			
City:	San Marcos	**State:**	TX
Zip Code:	78666-4684	**Congressional District:**	35
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	12/05/2016	**End Date:**	08/31/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rabenberg, Ellen	**Contact Phone:**	(256) 544-7264
Contact Email:	ellen.m.rabenberg@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 08/31/2026 per NSSC information (Ed., 10/10/25). NOTE: Per NASA-MSFC, the PI has received supplemental funding for an augmentation. End date changed to 08/31/2025 and grant number is now 80NSSC22K1361 (Ed., 4/1/25). NOTE: End date changed to 08/31/2024 per NSSC information (Ed., 9/17/23). NOTE: End date changed to 08/31/2022 per NSSC information (Ed., 9/6/22). NOTE: End date changed to 12/4/2021 per NSSC information (Ed., 3/10/21)		

Key Personnel Changes/Previous PI:	October 2025 report: Key Arizona State University (ASU) personnel on the original grant application are unchanged. This includes co-Principal Investigator (PI) Cheryl Nickerson and her team Jennifer Barrila (co-I) and Jiseon Yang (originally a postdoc, now Research Assistant Professor) at Arizona State University (ASU), as well as C Mark Ott (co-I) at NASA Johnson Space Center (NASA-JSC). Two additional participants, Sandhya Gangaraju and Richard Davis at Arizona State University (ASU) are also participating. Due to the expertise of our team, we performed all the necessary microscopy and will not require external collaborators as originally considered. In the PI's lab at Texas State (TxST), Starla Thornhill was a master's degree student as the grant was being written. She successfully defended her MS thesis in 2016 and completed her PhD in 2020 on work directly related to this grant. She continued as a postdoc during 2021 and along with Sandhya Gangaraju and Jiseon Yang (from ASU), completed all the postflight microscopy that flew on CRS-21. As well, we have added Alistair McLean as an unpaid collaborator. He works as a software engineer and wrote a program that helps us enumerate bacterial populations from microscopy images. With our re-flight occurring on CRS-29 in 2023, we added MS students Taylor Ranson (who did the confocal microscopy and culturing) and Evan Ortiz (who did the scanning electron microscopy and some computer image analysis). We also began a collaboration with Jelena Tesic (TxST computer science) and her PhD student Tanzina Acker Tani who did AI-enhanced image analysis of our microbial corrosion. Finally during the remainder of 2025-26, Peggy Biga (TxST Biology chair) will be serving as interim PI due to TxST university policy upon the formal retirement of Robert McLean. Robert McLean will still be involved and will be responsible for the scientific completion, manuscript writing, and any necessary reports.
COI Name (Institution):	Barrila, Jennifer Ph.D. (Arizona State University) Ott, C. Mark Ph.D. (NASA Johnson Space Center) Nickerson, Cheryl Ph.D. (Arizona State University)
Grant/Contract No.:	80NSSC22K1361 ; NNX17AC79G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Bacteria and other microorganisms have been shown to grow as surface-adherent biofilm communities in the microgravity environment during spaceflight. Organisms within biofilms are quite resistant to traditional antimicrobial chemicals and can foul water treatment filters resulting in potential risk to the astronauts. The biofilms on International Space Station (ISS) consist of many species, yet studies to date have only tested single-species biofilms. There is growing evidence that mixed-culture biofilms behave quite differently from pure culture biofilms, and we anticipate this phenomenon to occur in the microgravity environment of ISS. Through this study we will investigate the structure by which two-species, mixed-culture biofilms form under microgravity on stainless steel used to construct Environmental Controls and Life Support Systems (ECLSS) on the ISS. As well, we will explore the efficacy by which mixed-culture biofilms can be controlled, and whether microgravity-grown mixed-culture biofilms cause corrosion on stainless steel. As a result of supplemental funding, we are now able to gather genetic data on the mechanisms by which mixed bacterial species are able to form biofilms in microgravity conditions. Our results are designed to both enable future spaceflight missions and translate into information for the general scientific community on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	September 2023 Update: Our spaceflight experiment launched on SpaceX CRS-21 (SpX-21) on December 6, 2020 and two of our three samples returned on January 14, 2021. The third sample returned with Crew-1 on May 2, 2021. We encountered two major issues that have led us to request a reflight (which has been approved). One issue was media contamination, rendering our data interpretation as inconclusive. A second issue (needlestick injury to a space crewmember) resulted in some of our samples incubating for 117 days on the International Space Station (ISS) rather than the planned 4 days. While that caused a potential loss of some transcriptome data (described in the Task Progress section), a really positive development from the extended incubation resulted in enhanced microbial corrosion data, as well as the return of viable microorganisms that can now be investigated for genetic changes as a consequence of prolonged spaceflight. We are now pursuing an active study of experimental evolution in collaboration with the microbiology lab at NASA Johnson Space Center (JSC) (Co-I Mark Ott, Sarah Wallace, and their team). Another major positive development is the initiative to develop a machine learning computational approach to identify microbial-induced corrosion (MIC). MIC represents an estimated annual global economic loss of $500B and detection is a key need. We anticipate that our analytical approach in this area will have major economic benefits, far beyond the space program. October 2025 Update: Our re-flight on CRS-29 was quite successful and we are now in the process of completing our postflight analyses, and manuscript writing. We have submitted a conference paper and a separate manuscript on the machine learning computation to identify microbial induced corrosion on stainless steel; we observed a novel biofilm structure (analogous to Van Gogh's Starry Night) in our early time point (4d) spaceflight-grown biofilms; we observed silver resistance developing in spaceflight but not ground control cultures (genetic analysis is still ongoing); and we have biological duplicates of our extra-long time point samples (117d in spaceflight) on which we are doing genetic analysis to identify any unique mutation events that may occur. Finally all post-flight analysis and subsequent manuscript writing is in progress.
Task Progress:	Bacteria and other microorganisms often grow on surfaces as slimy layers, commonly referred to as biofilms. Two frequently encountered biofilms are dental plaque that grow on teeth, especially if tooth brushing is not done; and the slime that forms on shower walls and curtains if not cleaned. When growing as biofilms, bacteria are extremely difficult to control as they resist many disinfectants and other chemical treatments. In many industries, biofilms are associated with fouling and corrosion of pipelines and valves, heat exchangers, and corrosion caused by bacteria dissolving some metals by their metabolic activities. Biofilms also occur during spaceflight and represent a potential problem related to essential equipment, including life support systems. The Bacterial Adhesion and Control (BAC) study is geared towards studying biofilm formation by a mixture of two bacterial species, their ability to grow over time, their control by a silver-containing disinfectant and potential damage (corrosion) to stainless steel materials. We have found that the species composition and structure of biofilms changes over time and that space-grown biofilms form an interesting structure that resembles the famous "Starry Night" painting by Van Gogh. We are completing genetic analysis of our samples to understand the mechanisms that might explain these findings. We have developed a

	computer analysis program that allows the identification of microbial corrosion from electron microscopy images. Microbial corrosion is not unique to spaceflight, but represents a multi-billion dollar problem globally, so this new technology may have notable economic implications beyond the space program.
Bibliography Type:	Description: (Last Updated: 11/20/2025)
Articles in Peer-reviewed Journals	Akter Tani T, Scouten A, Ortiz E, McLean RJC, Tešić J. "Automated corrosion identification in metal imagery: Traditional vs. deep learning." Advances in Visual Computing. 2025 Jan. https://doi.org/10.1007/978-3-031-77389-1_21 , Jan-2025